

Muhammad Arsalan

Data Engineer | Apache Airflow · PySpark · AWS · Azure

marsalan8@outlook.com | +92 324 678 3480 | Karachi, Pakistan — Open to Remote (UTC+5)

linkedin.com/in/muhammad-arsalan-ai | github.com/Muhammad-Arsalan-Ai

SUMMARY

Data Engineer with 3+ years of experience building scalable production-grade data pipelines using Apache Airflow, PySpark, Kafka, and cloud platforms (AWS, Azure). Experienced in distributed data processing, workflow automation, real-time streaming, batch processing, and cloud-based data architecture. Proven track record delivering reliable ETL systems, optimizing pipeline performance, and maintaining high data quality across production systems.

SKILLS

Data engineering: Data Pipelines, ETL/ELT Pipelines, Apache Airflow, Data Orchestration, Workflow Automation, Apache Spark (PySpark, SparkSQL), Distributed Data Processing, Apache Kafka, Real-Time Data Streaming, Batch Processing, Data Modeling, Data Warehousing, Data Lake, Lakehouse Architecture, Delta Lake, Databricks, Snowflake, Medallion Architecture, OLAP (Online Analytical Processing), OLTP (Online Transaction Processing), Data Quality Monitoring, Data Governance

Languages & frameworks: Python, JavaScript, SQL, Flask, FastAPI, HTML, CSS, Bootstrap

Cloud & infrastructure: AWS (S3, Athena, Glue, EMR, Lambda, Kinesis, Redshift, RDS, EC2, IAM), Azure (ADF, Data Lake, Synapse Analytics), Cloud Data Architecture, Serverless Pipelines, Docker, Linux, CI/CD, Scalable Systems, Production Systems, Performance Optimization

Databases: PostgreSQL, MySQL, MongoDB (NoSQL), Qdrant (Vector DB), Redis

Data Analysis & Visualization: Power BI (DAX, OLAP), Streamlit, Matplotlib, Seaborn, Pandas, NumPy, Statistics

AI & LLMs: LangChain, RAG Pipelines, LLM Applications, Prompt Engineering, OpenAI API, Gemini, Hugging Face, Computer Vision

Tools & workflow: Git, Scrapy, Selenium, BeautifulSoup, Web Scraping, Jira, ClickUp, Excel, Google Sheets

PROFESSIONAL EXPERIENCE

Data Engineer

May 2025 – present

OfficeField

- Engineered Apache Airflow-orchestrated pipelines automating real-time stock and crypto data extraction from 5 financial APIs (FMP, Nasdaq, CoinMarketCap, CoinGecko ,etc), reducing data refresh cycles from daily to sub-hourly.
- Automated real-time ingestion and multi-destination storage across MongoDB, PostgreSQL, AWS RDS, and Qdrant Vector DB — processing thousands of financial records daily with zero manual intervention.
- Implemented automated data quality monitoring and validation checks across 4 production storage systems, achieving 99%+ data accuracy for downstream analytics consumers.
- Optimized workflow orchestration and Airflow task scheduling through parallelized ETL execution, improving pipeline performance and SLA compliance across real-time financial data pipelines.

Data Extraction Specialist (Freelancer)

Jul 2024 – Apr 2025

UpWork & Fiverr

- Delivered 30+ web scraping projects using Scrapy, Selenium, and BeautifulSoup — extracting and processing millions of records across paginated, dynamic, and CAPTCHA-protected websites.
- Automated end-to-end data collection workflows, reducing client manual effort by ~80% and delivering structured datasets within agreed SLAs on every project.

- Engineered robust web scraping solutions for JavaScript-rendered and dynamically paginated websites, maintaining 98%+ extraction completeness across high-volume datasets.

Data Engineer

Feb 2023 – Jun 2024

Xloop Digital Services (Pvt) Ltd

- Architected scalable distributed data processing pipelines using PySpark and AWS Glue to transform high-volume customer datasets powering real-time recommendation systems.
- Deployed a real-time Kafka streaming pipeline on AWS EC2 handling high-throughput event streams, integrated with AWS Glue for automated schema cataloging — reducing data availability lag from hours to under 5 minutes.
- Built a Python-based multi-database ETL pipeline to extract, clean, and segment customer data by region — loading USA records into MySQL and UK/India records into PostgreSQL, with structured JSON metadata for downstream processing.
- Designed DAX-optimized Power BI dashboards consolidating multi-source financial complaint datasets, enabling self-service analytics for 5+ stakeholders and eliminating recurring manual reporting.

PROJECTS

GovFinder Web Scraping

(Client Project — Private Repository)

Stack: Python, Scrapy, BeautifulSoup, PostgreSQL, CSV

- Developed an automated scraping tool to extract names, positions, and emails of American government officials from multiple official websites, centralizing data into a searchable structured database.
- Reduced government contact discovery time from hours of manual browsing to seconds of database querying.

Tamimi Markets Product Scraper

github.com/Muhammad-Arsalan-Ai/Tamimi-Markets-Product-Scraper

Stack: Python, Requests, CSV, REST API, JSON

- Engineered a scalable Python-based data extraction pipeline leveraging REST APIs to collect and normalize product data across 26 retail categories, enabling downstream analytics and reporting automation.
- Developed resilient extraction workflows with automated pagination, retry handling, and fault tolerance, improving dataset completeness and reducing scraping failures.

EDUCATION

Bachelor Of Computer Science

2019 – 2023

Sir Syed University of Engineering and Technology

CERTIFICATES

- Data Engineering Bootcamp — Emeritus · 2023
- Generative AI with Large Language Models — Coursera / DeepLearning.AI · 2024
- Certified Artificial Intelligence Developer — PIAIC · 2023
- Introduction to Data Analytics — LinkedIn Learning · 2024